

Missing data

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Recap: our course

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We are entering the last part of this course

1. R-skills, interpretation and non-linear effects
2. Data structures
 - ▶ Hierarchical data
3. Limited and categorical outcome variables (GLMs)
4. **Data structures** (today/last week, short)
 - ▶ **Missing data**

The purpose of this course

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⇒ *The purpose of this course is to find solutions when the assumptions of the linear model are not satisfied*

Two assumptions in ordinary regression

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Linear models (OLS) rely on two assumptions that are often violated

1. **Assumption 1:** outcomes are continuous and unbounded
2. **Assumption 2:** observations are independent and identically distributed
 - ▶ independent: probability of observing one unit is NOT dependent on observing another
 - ▶ identically distributed: observations come from the same data generating process/probability distribution

⇒ *strategies for when these are not satisfied*

Solutions to violations of those assumptions

1. **Assumption 1:** Limited and categorical outcome variables (GLMs):

- ▶ recode the dependent variable + probability distribution
- ▶ choice of model depends on the data generating process - e.g. logit, multinomial, ordinal, poisson, neg.bin, zero-inflated. . .

2. **Assumption 2:** Observations are not iid:

- ▶ hierarchical/nested data
- ▶ *missing data*

⇒ *what do we do when observations are not iid?*

Today

Sources of missing data

Sources of missing data

Missing data (NAs) are omni-present, and can create two problems

- ▶ statistical power: data set is too small (large SEs)
- ▶ biased sample: regression coefficients are wrong)

Discuss with your neighbour

- ▶ did anything chock you in the readings today? Why/why not?
- ▶ what are the common solutions to missing data?

Problem 1: statistical power

Two situations

- ▶ NAs are concentrated on some observations
 - ▶ is this systematic?
- ▶ NAs are spread out across variable
 - ▶ many small data leaks

Problem 2: selection bias

Most data contain missing observations

- ▶ missing data (NA) is the result of a “lurking” variable / process that :
 - ▶ assigns NA to some of the other variables
 - ▶ ... possibly affecting both x and y
- ▶ the “lurking” means that the assignment mechanism is not observed
 - ▶ think about the data generating process of the NA
 - ▶ we have to theorize / make assumptions

⇒ *missing observations might indicate a confounder*

What to do?

What to do?

$\Rightarrow$ *addressing/reducing the problem is often easier than what we think*

Classifications of missing data:

Take 1

The original classification by Rubin (1979)

- ▶ **MCAR** (Missing Completely at Random)
 - ▶ probability of NA is the same for all cases
- ▶ **MAR** (Missing at Random):
 - ▶ probability of NA depends on *observable* data (known sources)
- ▶ **MNAR** (Missing Not at Random)
 - ▶ probability of NA depends on *unobservable* data (unknown sources)

⇒ *these are assumptions that we can never test*

Why is it a problem ?

- ▶ **statistical power (MCAR):** only a problem if it reduces the N too much → *a representative sample*
- ▶ **information bias (MAR, MNAR):** we only record parts of a phenomenon (recall bias, misclassification, observer bias. . .)
 - ▶ independent variables:
 - ▶ we might not get the full “elasticity” of the variable
 - ▶ dependent variable: do we underestimate our phenomenon?
- ▶ **selection bias (MAR, MNAR):**
 - ▶ our estimate is biased because the unobserved assignment of NA affects both x and y

Take 2

We can subdivide the last category

- ▶ **MCAR** (Missing Completely at Random)
 - ▶ NA are *not* dependent on any predictors (observed or not): not conditional → *you can ignore the problem, unless you have too little statistical power*
- ▶ **MAR** (Missing at Random):
 - ▶ NA depends on the value of other observed predictors: it is conditional → *ignorability; you can condition on the other predictors*
- ▶ **MNAR** (Missing Not at Random)
 - ▶ NA depends on unobserved data
 - ▶ NA depends on the value of *the predictor itself* (e.g. censoring)

⇒ *model NAs to reduce bias in the estimate*

Discuss with your neighbour

Can you think of examples of missing data from the social sciences

1. identify a possible source of the missingness (DGP)
2. classify it as MCAR, MAR or MNAR

Strategies

Simple strategies

Discard data

Ignore the problem

- ▶ complete case analysis:
 - ▶ the usual “listwise exclusion”
- ▶ available data analysis:
 - ▶ analyze subsets of data separately
 - ▶ exclude variables with missing observations
- ▶ weighing of NA according to predictors
 - ▶ common in surveys → *some cases may be underrepresented in the data, because of NA*

Replace each NA by a single value

We can also infer the missing values in fairly simple ways

▶ **mean imputation:**

- ▶ replace the missing data by the variable mean

▶ **conditional mean imputation:** use information from other variables

- ▶ group mean, regression predictions

⇒ *doesn't take into account the uncertainty from our estimate*

Multiple imputations

Multiple imputations

Multiple imputation generates several predictions for each missing value to account for the uncertainty

- ▶ step 1: make predictions for the missing values by adding some random noise for each model

→ *we end up with several data sets (5-20 frames)*

- ▶ step 2: estimate the main model on all the different data sets

→ *pool over the regression parameters*

EM algorithm

EM algorithm

The EM is the base-line approach, and only has one data frame in the end

- ▶ we have several variables
- ▶ E-step:
 - ▶ give your NA some initial values
 - ▶ predict your x_{miss} using the observed values and initial place-holder values for all other variables
- ▶ M-step:
 - ▶ re-do until you your predictions of x_{miss} don't change any more (set a value at which you stop)

⇒ *one best estimand, no uncertainty*

Multiple Imputation via Chained Equations (MICE)

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We assume a set of variables are correlated, and use them to predict for each other in turn (a cycle)

Imagine x , y and z :

- ▶ cycle 1:
 - ▶ $x \propto \alpha + \beta_1 y + \beta_2 z$: give y and z some starting value; regress x on all other models
 - ▶ $y \propto \alpha + \beta_1 \hat{x} + \beta_2 z$: replace missing values of x by predicted $\hat{x}$
 - ▶ $z \propto \alpha + \beta_1 \hat{x} + \beta_2 \hat{y}$: same
- ▶ cycle 2-...: rinse and repeat until nothing changes (convergence)



Figure 1: Mice thrive in holes...

⇒ *each model is separate / can be a separate GLM*

Discuss with your neighbour

What do you think about:

- ▶ using y to impute x ?
- ▶ using x to impute y ?
- ▶ using missing observations to impute x or y ?
- ▶ using only variables *in* the main model?
- ▶ selection criteria for using other variables?

⇒ *any other objections?*

Two risks when you impute

Two risks when you impute

- ▶ over-confidence: estimate the uncertainty from the imputation
- ▶ increased / continued bias: placebo tests

Add som random noise

This is usually done together with a bit of (random) noise at each step

- ▶ for each iteration, create a new data set with imputed variables
- ▶ estimate your main model on each data set:
 - ▶ regression parameters (β) are averaged over
 - ▶ the standard error is a fusion of both:
 - ▶ within-model variation: standard errors from the regression
 - ▶ between-model variation: the deviation between the regression parameters

...then check

... were my imputations appropriate?

- ▶ problem of **overfitting**:
 - ▶ you may have perfect in-sample predictions, that are useless for out-of-sample imputation
- ▶ **overimputation**: randomly leave observations out, check if you predict correctly

Takeaway

- ▶ three problems with missing values:
 - ▶ low N
 - ▶ partial x / y
 - ▶ selection bias
- ▶ solutions
 - ▶ listwise exclusion
 - ▶ weights
 - ▶ imputations

Workflow (R-notebook)

- ▶ Map the missing:
 - ▶ how many / which variables?
 - ▶ do they overlap?
 - ▶ do NAs correlate with observables?
- ▶ Diagnose: MCAR; MAR; MNAR
- ▶ Address: often multiple imputation
- ▶ Check results: does it hold up?

Literature